



UPI QR Code Payment

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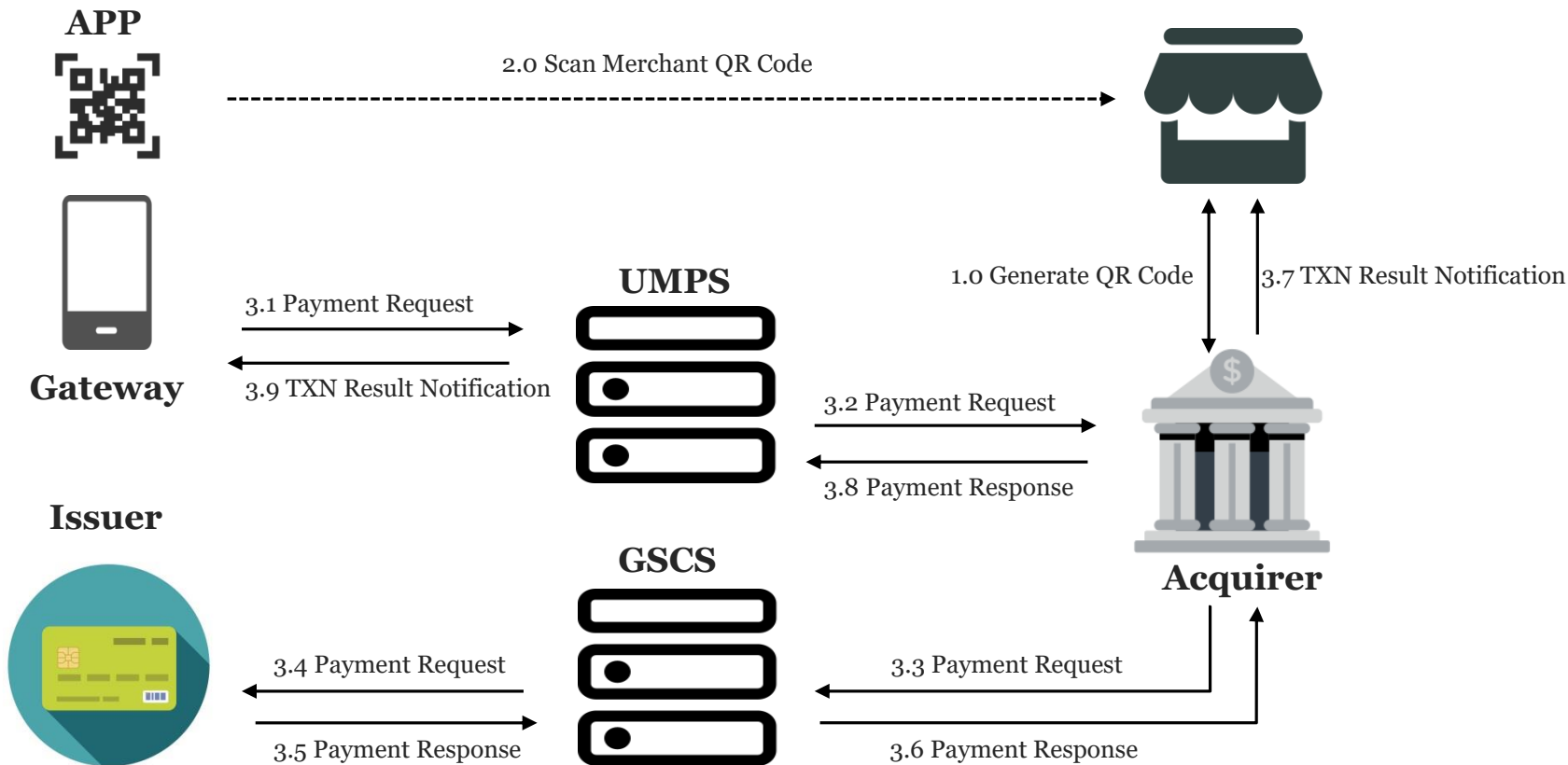
Contents

- **QR Code Payment Transaction**
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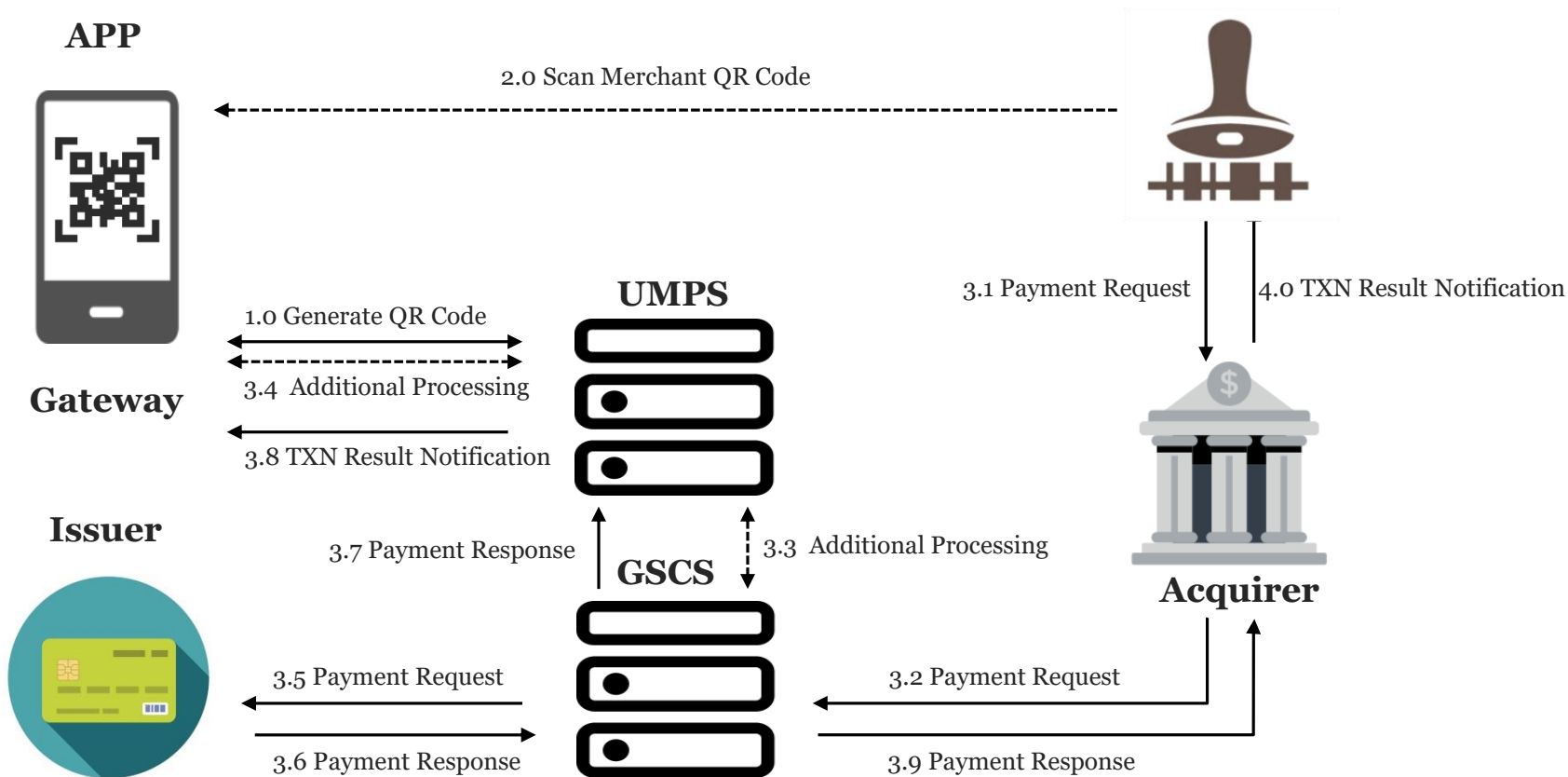
01

QR Code Payment Transaction

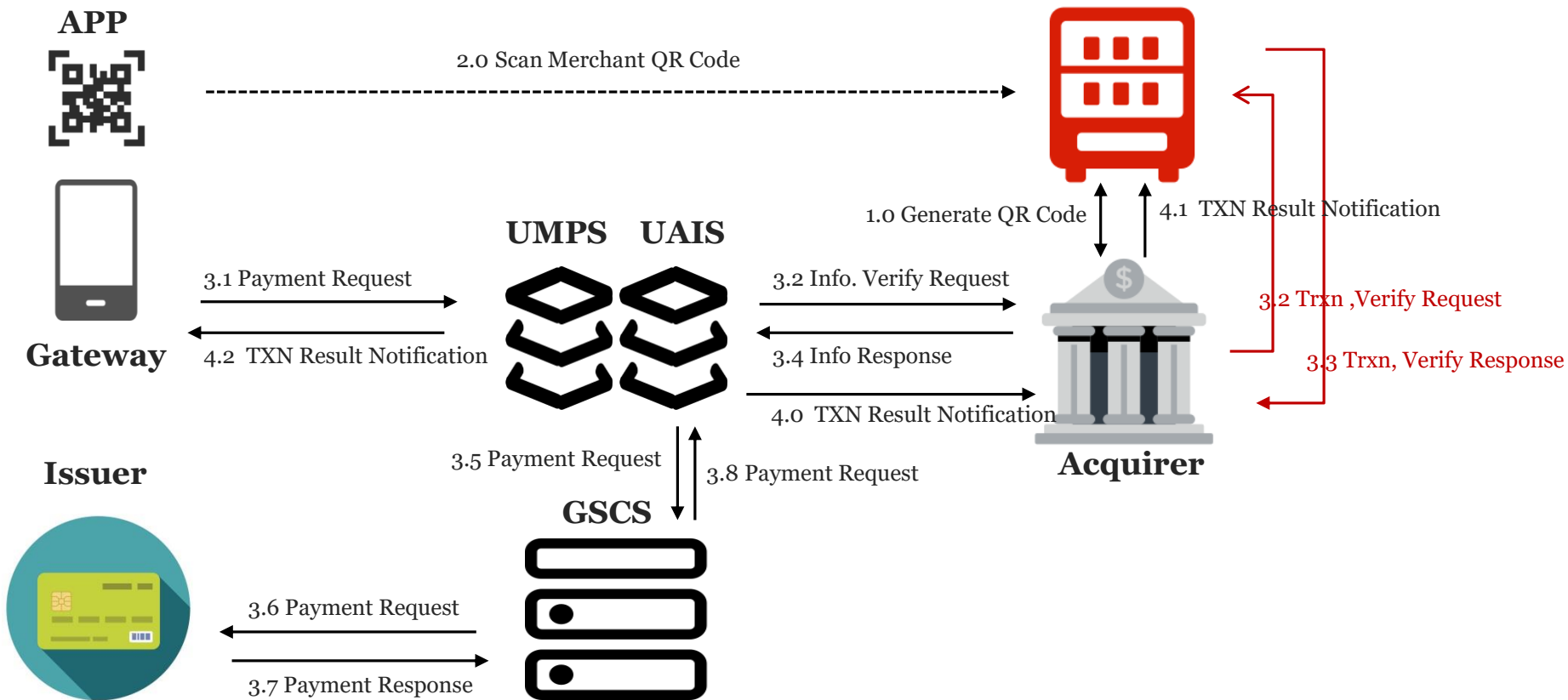
UPI QR Code - Merchant Presented (ISO 8583)



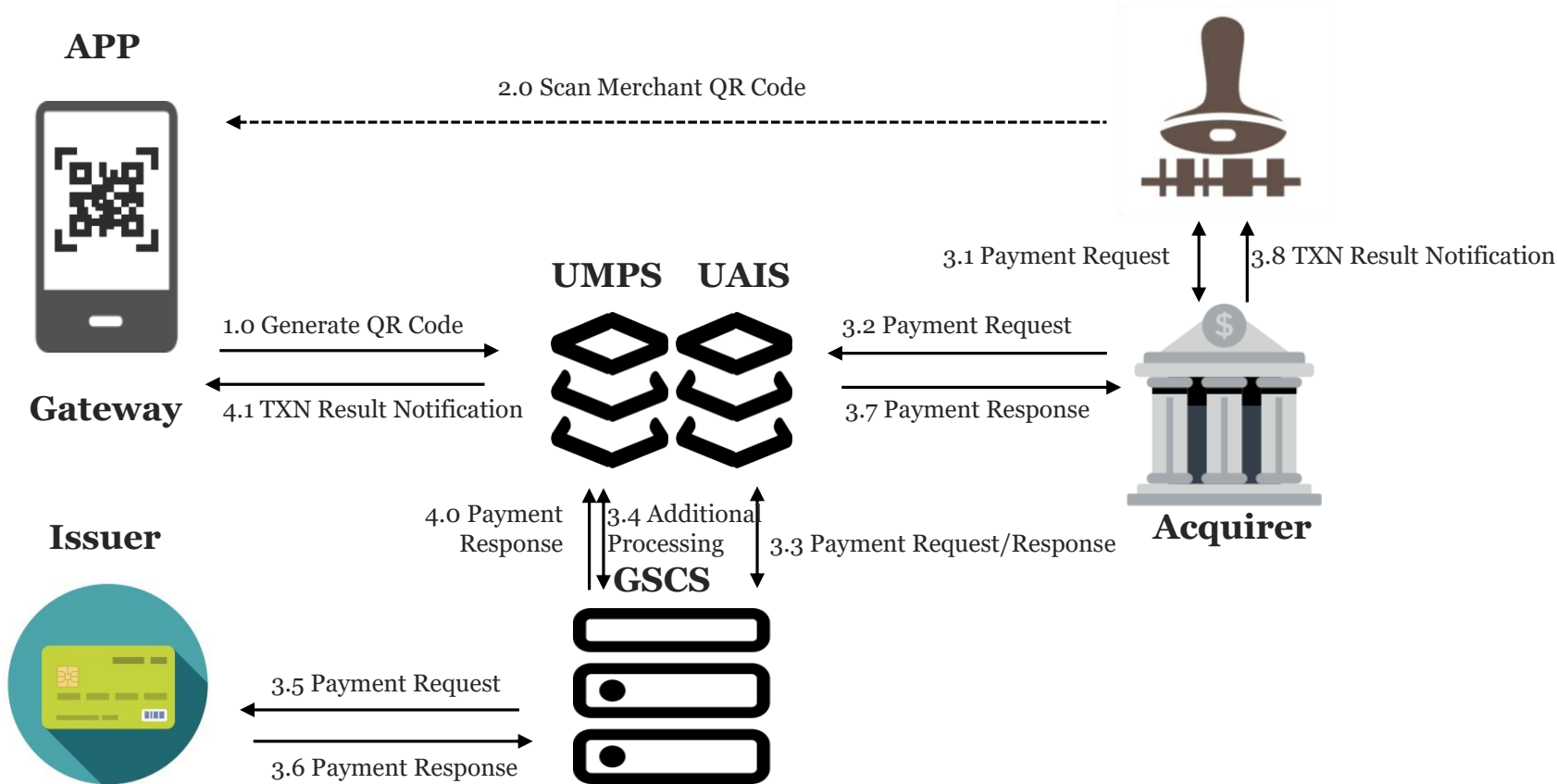
UPI QR Code - Consumer Presented (ISO 8583)



UPI QR Code - Merchant Presented (JSON)



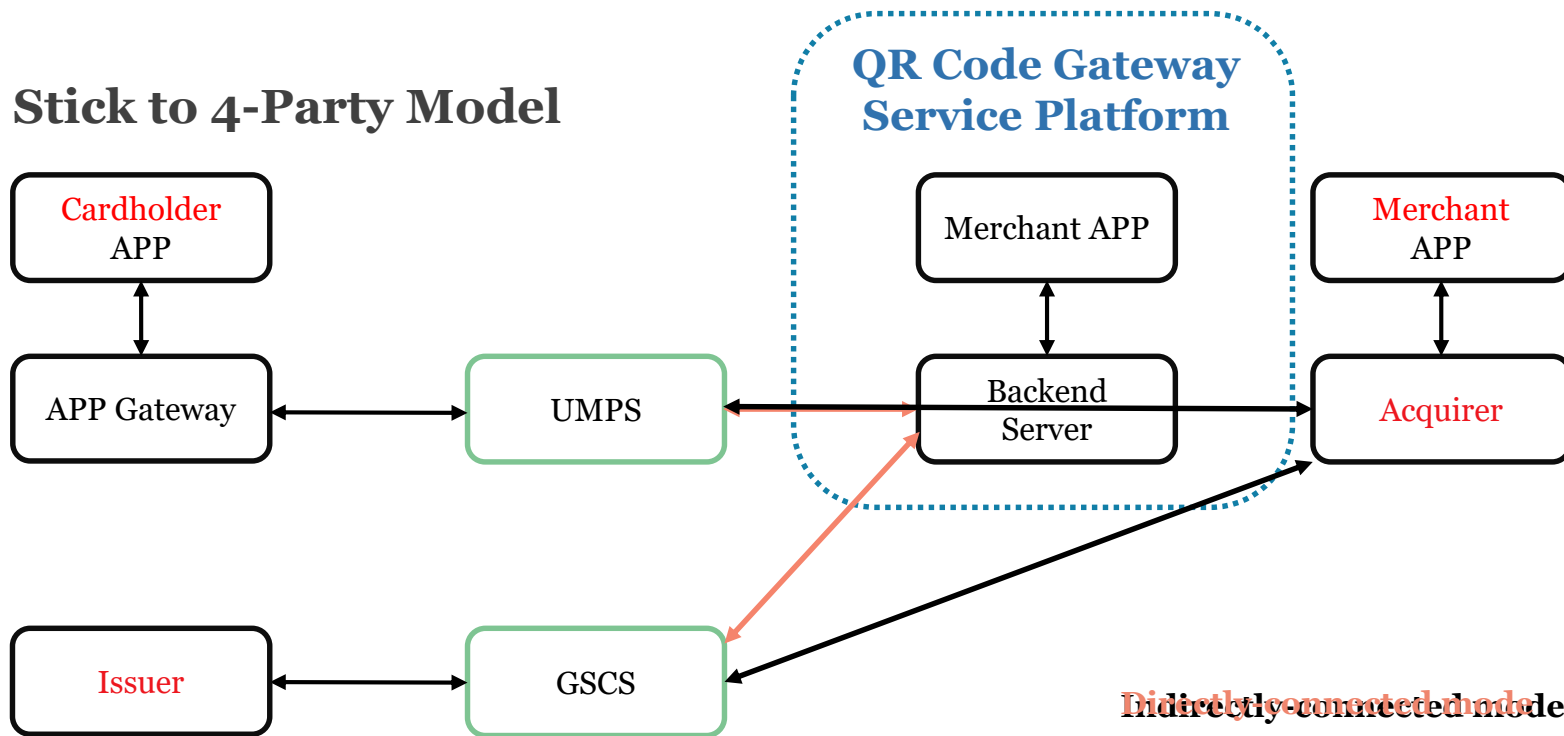
UPI QR Code - Consumer Presented (JSON)



QR Code Gateway Service Platform



Stick to 4-Party Model



02

System Implementation

Implementation of Issuer



- ❑ Online Message and File Interface
- ❑ Account Verification
- ❑ Payment Authorization
- ❑ Cancellation
- ❑ Refund
- ❑ Reversal



- ❑ Card/Account Verification
- ❑ Consumer & Merchant-Presented
- ❑ EMV & URL Request and Decode
- ❑ Additional Processing
- ❑ Transaction Notification
- ❑ Transaction Inquiry

Implementation of Acquirer



Acquirer Host System

- ❑ Online Message and File Interface
- ❑ Payment Transactions
- ❑ Transaction Status Inquiry
- ❑ Cancellation and Refund using QRC voucher number

Merchant Management Server

- ❑ Merchant Enrollment
- ❑ QR Code generation and management (Static/Dynamic)
- ❑ Merchant Payment Notification
- ❑ Cancellation/Refund using QRC voucher number
- ❑ Statistics and Reporting

Merchant Client

POS Terminal, Merchant MIS, Merchant APP, E-Commerce Website, TV E-Commerce System, Multi-media Terminals, Vending Machines etc.

Product Implementation Guide



Acquirer



Issuer



- ✓ Independent QRC acquiring license
 - ✓ acquirers that obtained POS license already can automatically enrolled with QRC license
- ✓ Pull Mode for Acquirer : Estimated 2- 3 months
- ✓ Push Mode for Acquirer : Estimated 2 - 3 months

- ✓ For Issuer : Estimated 3 – 4 months
- ✓ With UPI QRC API, the implementation time can be largely reduced

Product Implementation Guide



Specification	Target	Description
UPI QR Code Encoding Specification	Acquirer/Issuer	The technical specification to help UPI partners generate or parse the QR Code for consumer and merchants following EMVCo standard
UPI Technical Specifications on Bankcard Interoperability	Acquirer/Issuer	The technical specification to explain data elements required by UPI system (UMPS) to transit UPI QR Code transaction.
UnionPay Mobile Payment Service Technical Specifications	Issuer/APP Provider	The technical specification to help issuer or APP Provider develop the mobile APP and connect with UnionPay system
UnionPay Interface Specification of CUP QR and UMPS	Issuer/APP Provider	The technical specification to help issuer or APP Provider from mainland of China connect the mobile APP with UnionPay system for cross-border QR Code business
UPI QR Code Payment (ISO 8583) Implementation Guide	Acquirer/Issuer	The documents describes necessary changes of Partners to support QR Code payments through ISO 8583, including requirements for network access, the transaction process etc.

Product Implementation Guide



Specification	Target	Description
UPI Acquiring Interface Service Technical Specifications	Acquirer	The technical specification to explain data elements required by UPI system (UAIS) to transit QR Code transactions from Acquirers with JSON interface
UPI QR Code Payment (JSON Interface) Implementation Guide	Acquirer	The documents describes necessary changes of Acquirer to support QR Code payments with JSON interface, including requirements for network access, the transaction process etc.
UPI Application Specification for Direct-Connected POS Terminal	Acquirer	The technical specification for Acquirers to modify the POS terminal and connect terminal directly to UnionPay system for QR Code
UPI QR Code Payment Implementation Guide for Direct-Connection	Acquirer	The documents describes necessary procedures for Acquirer to support QR Code payments with direct connection of merchants to UPI system
UPI QR Code Testing Case Guide	Acquirer/Issuer/ APP Provider	The test cases for partners to examine the system development status for QR Code transaction under different scenarios.

Thanks

Cooperation

Efficiency

Creativity

Impact to Issuer System – Online Message



Transaction Type	Online Message (ISO 8583) UPI to Issuer					
	F22	F125	60.2.5	60.3.6	F23 F35 F55	Others
Purchase/Authorization	04: Consumer presented QR code without chip data; 94: Merchant presented QR code(transaction without chip data);	Conditional *F125 is used to transfer additional info of QRC transaction (use case ID, App provider IIN, etc.)	03(Consumer presented) 07(Merchant presented)	Field 60.3.6 will be filled with 5 to indicate a QRC transaction	Issuer generated EMV QRC transaction: Will be present Other Scenarios : Will not be present <i>Other scenarios includes: Issuer generated barcode transaction, Merchant presented mode transaction.</i>	Same as POS Transaction
Reversal	Same as original transaction	As above	As above	As above	Will not be present	
Cancellation	04: Consumer presented QR code without chip data; 94: Merchant presented QR code(transaction without chip data);	As above	As above	As above		
Refund	04: Consumer presented QR code without chip data; 94: Merchant presented QR code(transaction without chip data);	As above	As above	As above		

Impact to Issuer System – Online Message



Transaction Type	Online Message (ISO 8583) Issuer to UPI					
	F22	F125	60.2.5	60.3.6	F23 F35 F55	Others
Purchase/Authorization	Same as request message					
Reversal					Same as request message	Same as POS Transaction
Cancellation						
Refund						

Impact to Issuer Host System – File Interface



File Interface

Consumer Presented Mode: Valid value(04) needs to be identified in Field 22.

Merchant Presented Mode: Valid value(94) needs to be identified in Field 22.

In the presentment file and file refund, if **QRC Voucher Number** is supported, it's filled in the Block 1 position 91-110 of TC100

Appendix-Impact to Acquirer Host System – Online Message



Transaction Type	Online Message (ISO 8583)														
	UMPS->ACQ (Message Type ID:0200/0100)								ACQ->UMPS (Message Type ID:0210/0110)						
	F2	F22	F60.2.5	F60.3.6	F63 MQ	F125	F113*	Other	F2	F22	F60.2.5	F60.3.6	F63	F125	Other
Purchase	M	94	07	/	C	M	C	Same as POS transaction	M	94	07	/	/	->	Same as POS transaction
Reversal	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
Cancellation	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
Refund	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/

Transaction Type	Online Message(ISO 8583)											
	ACQ->GSCS (Message Type ID:0200/0100)						GSCS->ACQ (Message Type ID:0210/0110)					
	F2	F22	F60.2.5	F60.3.6	F125	Other	F2	F22	F60.2.5	F60.3.6	F125	Other
Purchase	M	94	07	/	M	Same as POS transaction	M	94	07	->	M	Same as POS transaction
Reversal	M	94	07	/	M	Same as POS transaction	M	94	07	->	M	Same as POS transaction
Cancellation	M	94	07	/	M	Same as POS transaction	M	94	07	->	M	Same as POS transaction
Refund	M	94	07	/	M	Same as POS transaction	M	94	07	->	M	Same as POS transaction

* If ACQ support UPI UPLAN, ACQ shall support Field 113 to get UPLAN discount information.

Impact to Acquirer System – Online Message



Transaction Type		Online Message (ISO 8583) UPI (GSCS) -> ACQ		Online Message (ISO 8583) ACQ -> UPI (GSCS)						
		F125 (QRC voucher Number)	Other	F2	F22	F23	F35	F55	F125	Other
Purchase (Single Message) /Authorization Transaction with Fixed Amount(Dual Message)		Conditional for Acquirer to Receive	Same as IC Card Transacti on	Terminal retrieve from Tag 57 in QR Code	F22: 03 (Consumer presented QR Code, chip information included)	Terminal retrieve Tag 5F34 from QR Code. Pad one leading zero And Fill into F23 E.g. Tag 5F34=01, fill into F23 with 001)	Tag 57 retrieved by terminal from QR Code	Include : 1. Card data Terminal retrieve from Tag 63 (including all sub Tags) in QR Code. 2. Terminal data use default values for terminal data	/	Same as Chip Transacti on
Reversal		Optional for Acquirer to Receive					NOT Required for Acquirer to Send		/	
Cancell ation	If Transaction Receipt Supported on terminal	Optional for Acquirer to Receive					Conditional (this field appear if tag57 is used in the transaction)	NOT Required for Acquirer to Send	Optional for Acquirer to Send	
	If Transaction Receipt NOT Supported on terminal	Required for Acquirer to receive							Optional for Acquirer to Send	
Refund	If Transaction Receipt Supported on terminal	Optional for Acquirer to Receive							Optional for Acquirer to Send	
	If Transaction Receipt NOT Supported on terminal	Required for Acquirer to Receive							Required for Acquirer to Send	

Notes: QRC voucher number is a UnionPay generated unique number that can be used to locate a UnionPay QRC transaction

Impact to Terminals



Transaction Type		Impact to Terminals	
		Terminal Parameter	Terminal Function
Purchase(Single Message)/Authorization Transaction with Fixed Amount(Dual Message)		Terminal retrieves Tag 4F in 61 template from QR Code. If got UnionPay AID, terminal continues with subsequent processing.	Same as traditional POS. Optional for receipt printing.
Reversal			Same as traditional POS.
Cancellation	If Transaction Receipt Supported on terminal		1.Cancellation can be initiated with traditional Transaction Trace Number at terminal end. 2.Cancellation can also be initiated with QR Code Voucher Number at terminal end. (Optional)
	If Transaction Receipt NOT Supported		Cancellation is required to be initiated with QR Code Voucher Number at terminal end. QRC Voucher number can be provided by cardholder from APP transaction notification.
Refund	If Transaction Receipt Supported on terminal		1.Refund can be initiated with scanning QR Code + original reference number + refund amount + original transaction date at terminal end. 2.Refund can also be initiated with QR Code Voucher Number at terminal end. (Optional)
	If Transaction Receipt NOT Supported		Refund is required to be initiated with QR Code Voucher Number at terminal end. QRC Voucher number can be provided by cardholder from APP transaction notification.

Impact to Acquirer Host System – File Interface



File Interface

Block 1 is required in outgoing file, valid value(94) needs to be identified in Field 22.

QRC Voucher Number is required for refund for dual message institutions, use position 91-110 in Block 1 for QRC Voucher Number.